

# Chaehyeong Lee

Boulder, Colorado, USA

+1 (303) 258-6841 • Chaehyeong.Lee@colorado.edu

sites.google.com/yonsei.ac.kr/hyeong •  https://orcid.org/0009-0005-3110-9839

## Research Interests

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Geofluid Dynamics: Cross-scale upper-ocean mixing processes; Mesoscale eddies.

Physical Oceanography: Long-term change in the upper-ocean thermal state; Ocean's role in climate systems.

## Education

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**University of Colorado Boulder**

*Ph.D. in Atmospheric and Oceanic Sciences*

Advisor: Dr. Donata Giglio & Co-Advisor: Dr. Aneesh Subramanian

**Boulder, CO, USA**

*Aug. 2024 – present*

**Yonsei University**

*M.S. in Atmospheric Sciences*

Advisor: Dr. Hajoong Song

**Seoul, Rep. of Korea**

*Aug. 2023*

**Yonsei University**

*B.S. in Atmospheric Sciences*

Advisor: Dr. Hajoong Song

**Seoul, Rep. of Korea**

*Feb. 2022*

## Publications

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Under Review.....

1. Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., Molod, A., & Ruiz-Xomochuk, V. (2026). Effects of Satellite Sea Surface Salinity Assimilation on the Representation of Upper-Ocean Thermohaline Structure and Variability in the NASA GEOS-S2S-2 model. *Quarterly Journal of the Royal Meteorological Society*

In Preparation.....

1. Lee, C., Giglio, D., Subramanian, A. C., Mazloff, M., & Arbic, B. K.

Published.....

1. Lee, C., Song, H., Choi, Y., Cho, A., & Marshall, J. (2025). Observed Multi-decadal Increase in the Surface Ocean's Thermal Inertia. *Nature Climate Change*, **15**, 308-314. doi:10.1038/s41558-025-02245-w

## Research Experience

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**Dr. Giglio's Research Group, CU Boulder**

*Research Assistant*

**Boulder, CO, USA**

*Aug. 2024 – present*

**Climate Processes and Predictability Group, CU Boulder**

*Research Assistant*

**Boulder, CO, USA**

*Aug. 2024 – present*

**Air-Sea Modeling Lab, Yonsei University**

*Research Assistant*

**Seoul, Rep. of Korea**

*Dec. 2020 – Aug. 2024*

## Teaching Experience

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|----------------------------------------------------------------|---------------------------------------------------|
| <b>Data Science Laboratory</b><br><i>Teaching Assistant</i>    | <b>Boulder, CO, USA</b><br><i>Spring 2026</i>     |
| <b>Climate &amp; Civilization</b><br><i>Teaching Assistant</i> | <b>Seoul, Rep. of Korea</b><br><i>Spring 2023</i> |
| <b>Physical Oceanography</b><br><i>Teaching Assistant</i>      | <b>Seoul, Rep. of Korea</b><br><i>Fall 2022</i>   |

## Awards & Scholarships

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**2025:** ATOC Fellowship, Department of Atmospheric and Oceanic Sciences, University of Colorado Boulder  
**2025:** Academic Research Grants (GCP research credits), Google LLC  
**2024:** Outstanding Thesis Award, Yonsei University Graduate School, Yonsei University  
**2022–2023:** Full tuition merit scholarship (for the top 2 graduate students), Yonsei University  
**2022:** High Honors for academic performance, Yonsei University  
**2020–2021:** Jilli Scholarship for academic performance, Yonsei University

## Patent

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Song, H., & Lee, C. (2025). *Evaluation system and method of persistence of SST anomalies using autocorrelation coefficient and arctangent regressive model*. Rep. of Korea Patent #1028135790000. doi:10.8080/1020220157159

## Invited Talk

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| <b>NASA Salinity Telecon</b><br><i>Assessing the Impact of Satellite Sea Surface Salinity Assimilation on Vertical Structure of the Upper Ocean in the NASA GEOS-S2S 2.</i> | <b>Virtual Meeting</b><br><i>Dec. 2025</i> |
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## Selected Presentations

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| <b>2027 American Meteorological Society Annual Meeting (poster)</b><br><i>Ocean Advection Drives Mixed Layer Temperature Variance at 10-100 km in the Gulf Stream Extension.</i><br><i>Lee, C., Giglio, D., &amp; Subramanian, A. C.</i>                                                     | <b>Denver, CO, USA</b><br><i>Jan. 2027 (scheduled)</i>  |
| <b>US Climate Variability and Predictability (poster)</b><br><i>Enhanced Upper-Ocean Horizontal Advection in a High-Resolution Climate Model Improves the Representation of High-Frequency Near-Surface Temperature Variability.</i><br><i>Lee, C., Giglio, D., &amp; Subramanian, A. C.</i> | <b>Boulder, CO, USA</b><br><i>Sep. 2026 (scheduled)</i> |
| <b>Ocean Science Meeting 2026 (poster)</b><br><i>Assessing the Impact of Satellite Sea Surface Salinity Assimilation on the Upper Ocean Thermal State in the NASA GEOS S2S-v2 Model.</i><br><i>Lee, C., Giglio, D., Subramanian, A. C., Han, W., Capotondi, A., Du, D., &amp; Molod, A.</i>  | <b>Glasgow, Scotland</b><br><i>Feb. 2026</i>            |

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| <b>American Geophysical Union Fall Meeting (poster)</b><br><i>The Increasing Trend of Persistence of Sea Surface Temperature in the Past 40 Years.</i><br><i>Lee, C., Song, H., Cho, A., &amp; Tak, Y.</i> | <b>Chicago, IL, USA</b><br><i>Dec. 2022</i>    |
| <b>Korean Society of Oceanography Spring Conference (talk)</b><br><i>Increasing Persistence of SST Anomalies and Duration of Marine Heatwaves.</i><br><i>Lee, C., Song, H., Cho, A., &amp; Tak, Y.</i>     | <b>Jeju, Rep. of Korea</b><br><i>Jun. 2022</i> |

## Workshops

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| <b>Broadening Earth System Research Across Scales with High-Resolution Modeling Workshop</b><br><i>US Climate Variability and Predictability</i>  | <b>Boulder, CO, USA</b><br><i>Sep. 2026 (scheduled)</i> |
| <b>Data Assimilation Training and Science Workshop</b><br><i>Consortium for Advanced Data Assimilation Research and Education &amp; NOAA EPIC</i> | <b>Norman, OK, USA</b><br><i>Jun. 2026</i>              |
| <b>14th International Workshop on Modeling the Ocean 2024</b><br><i>International Workshop on Modeling the Ocean</i>                              | <b>Sapporo, Japan</b><br><i>Jun. 2024</i>               |
| <b>User Training for the Glosea 6 Climate Prediction Model</b><br><i>Korea Meteorological Administration</i>                                      | <b>Jeju, Rep. of Korea</b><br><i>Jan. 2022</i>          |
| <b>Deep Learning Training: Fundamentals of Deep Learning</b><br><i>NVIDIA Deep Learning Institute</i>                                             | <b>Gonju, Rep. of Korea</b><br><i>Jan. 2022</i>         |

## Academic Service

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Peer Reviewer: *Journal of Climate*

## Technical Skills

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**Programming:** Python (xarray, dask, Pangeo), Julia

**Modeling:** MITgcm, Oceananigans

**HPC:** Parallel/distributed computing, NCAR Casper/Derecho clusters

**Tools:** Git, Linux shell scripting, LaTeX